

National Income: How It Is Spent

Chapter 4

Econ 313 – Intermediate Macroeconomics

University of Tennessee, Knoxville

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In This Chapter, You Will Learn:

- ▶ How total income is distributed
- ▶ What determines the demand for goods and services
- ▶ How equilibrium in the goods market is achieved

Demand for goods and services (1 of 2)

- ▶ Demand side
 - ▶ determinants of the four components of GDP:
 - ▶ C — *consumption*
 - ▶ I — *investment*
 - ▶ G — *government expenditures*
 - ▶ NX — *net exports*

Demand for goods and services (2 of 2)

Components of aggregate demand:

C = consumer demand for goods and services

I = demand for investment goods

G = government demand for goods and services

(closed economy: no NX)

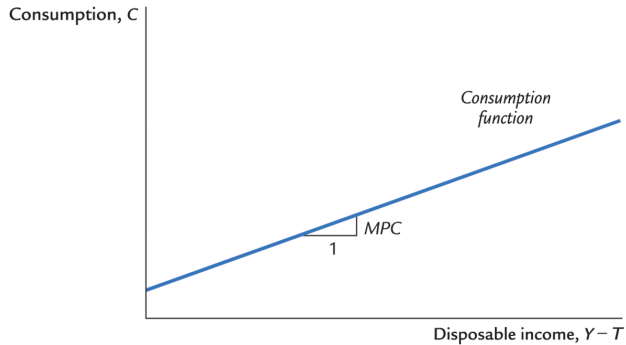
Consumption, C (1 of 2)

- ▶ Households receive income from their labor and from owning capital.
- ▶ They pay taxes to the government.
- ▶ They decide how much of their after-tax income to consume and save.
- ▶ How much to consume and save (and how much to work) are the typical decisions households make in many economic models.

Consumption, C (2 of 2)

- ▶ **Disposable income** is total income minus total taxes: $Y - T$.
- ▶ Consumption function: $C = C(Y - T)$
- ▶ Definition: **marginal propensity to consume (MPC)** is the change in C when disposable income increases by one dollar.

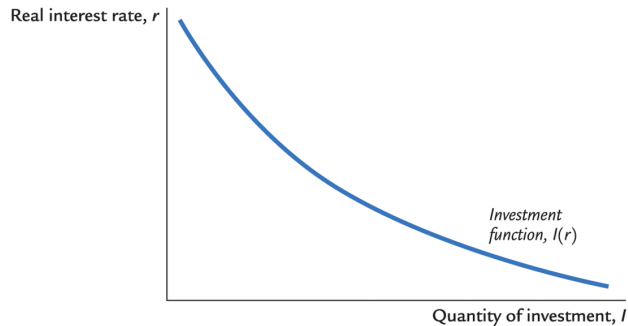
The consumption function



Investment, I

- ▶ The investment function is $I = I(r)$, where r denotes the **real interest rate**, the nominal interest rate corrected for inflation.
- ▶ The real interest rate is:
 - ▶ the cost of borrowing
 - ▶ the opportunity cost of using one's own funds to finance investment spending
- ▶ So, I depends negatively on r :
 - ▶ Here, r is the real interest rate, not the rental rate of capital (R).

The investment function



Government spending, G

- ▶ G = government spending on goods and services
- ▶ G excludes transfer payments (for example, Social Security benefits, unemployment insurance benefits)
- ▶ Assume that government spending and total taxes are exogenous:

$$G = \bar{G} \text{ and } T = \bar{T}$$



U.S. Capitol — Congress appropriates $\bar{G}$ and sets $\bar{T}$.

Wikimedia, PD

The market for goods and services

Aggregate demand: $C(\bar{Y} - \bar{T}) + I(r) + \bar{G}$

Aggregate supply: $\bar{Y} = F(\bar{K}, \bar{L})$

Equilibrium: $\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}$

*The real interest rate adjusts
to equate demand with supply.*

The loanable funds market

- ▶ A simple supply–demand model of the financial system.
- ▶ One asset: “loanable funds”
 - ▶ demand for funds: investment
 - ▶ supply of funds: saving
 - ▶ “price” of funds: real interest rate

Demand for funds: Investment

The demand for loanable funds:

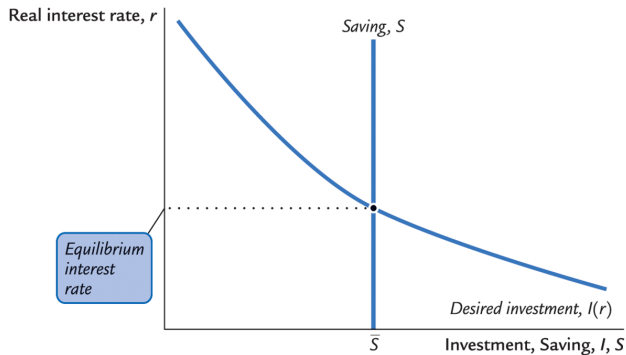
- ▶ comes from investment:

Firms borrow to finance spending on plant and equipment, new office buildings, etc. Consumers borrow to buy new houses.

- ▶ depends negatively on r :

r is the “price” of loanable funds (cost of borrowing).

Loanable funds demand curve



Supply of funds: Saving

- ▶ The supply of loanable funds comes from saving:
 - ▶ Households use their saving to make bank deposits and purchase bonds and other assets. These funds become available to firms to borrow and finance investment spending.
 - ▶ The government may also contribute to saving if it does not spend all the tax revenue it receives.

Types of saving

Private saving $= (Y - T) - C$

Public saving $= T - G$

National saving, S

$= \text{private saving} + \text{public saving}$

$= (Y - T) - C + T - G$

$= Y - C - G$

Notation: Δ = change in a variable

For any variable $\mathbf{X}$, ΔX = “change in X ”
 Δ is the Greek (uppercase) letter *delta*.

Examples:

- ▶ If $\Delta L = 1$ and $\Delta K = 0$, then $\Delta Y = MPL$, as $Y = F(K, L)$.

More generally, if $\Delta K = 0$, then: $MPL = \frac{\Delta Y}{\Delta L}$.

- ▶ $\Delta(Y - T) = \Delta Y - \Delta T$, so
 $\Delta C = MPC \times (\Delta Y - \Delta T)$
 $= MPC \Delta Y - MPC \Delta T$

NOW YOU TRY: Calculate the change in saving

Suppose $MPC = 0.8$ and $MPL = 20$.

For each of the following, compute ΔS :

(a) $\Delta G = 100$

(b) $\Delta T = 100$

(c) $\Delta Y = 100$

(d) $\Delta L = 10$

NOW YOU TRY: Calculate the change in saving, answers

$$\begin{aligned}\Delta S &= \Delta Y - \Delta C - \Delta G = \Delta Y - 0.8(\Delta Y - \Delta T) - \Delta G \\ &= 0.2\Delta Y + 0.8\Delta T - \Delta G\end{aligned}$$

(a) $\Delta S = -100$

(b) $\Delta S = 0.8 \times 100 = 80$

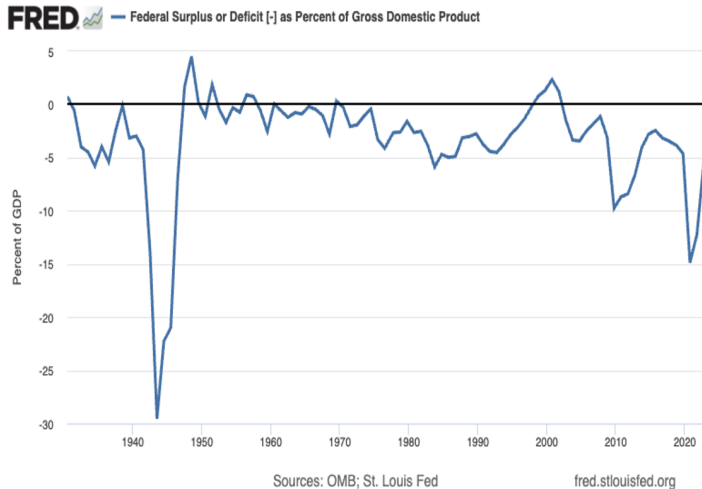
(c) $\Delta S = 0.2 \times 100 = 20$

(d) $\Delta Y = MPL \times \Delta L = 20 \times 10 = 200,$
 $\Delta S = 0.2 \times \Delta Y = 0.2 \times 200 = 40.$

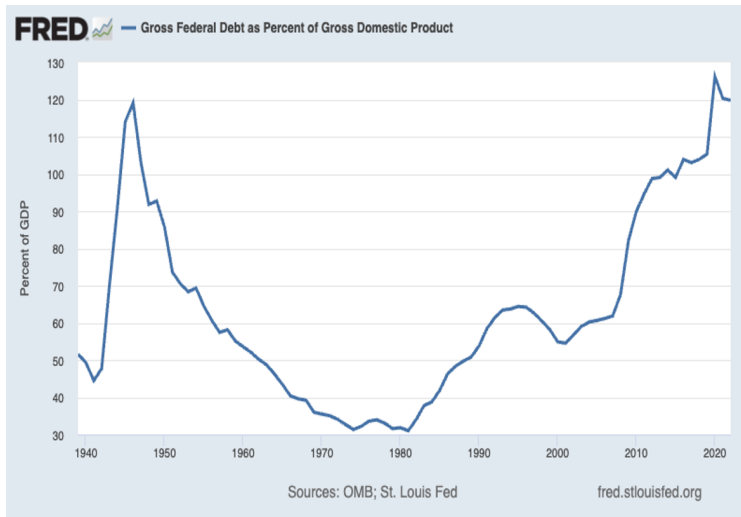
Budget surpluses and deficits

- ▶ If $T > G$, **budget surplus** $= (T - G)$ = public saving.
- ▶ If $T < G$, **budget deficit** $= (G - T)$ and public saving is negative.
- ▶ If $T = G$, **balanced budget**, public saving $= 0$.
- ▶ The U.S. government finances its deficit by issuing Treasury bonds—that is, borrowing.

U.S. federal government surplus/deficit, 1930–2022



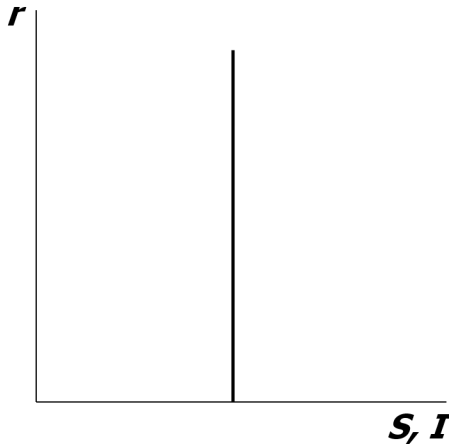
U.S. federal government debt, 1940–2022



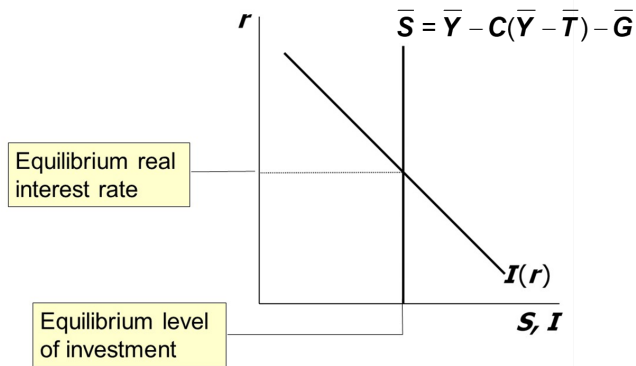
Loanable funds supply curve

National saving does not depend on r , so the supply curve is vertical.

$$\bar{S} = \bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G}$$



Loanable funds market equilibrium



The special role of r

r adjusts to equilibrate the goods market *and* the loanable funds market simultaneously:

If the loanable funds market is in equilibrium, then

$$Y - C - G = I$$

Add $(C + G)$ to both sides to get

$$Y = C + I + G \quad (\text{goods market equilibrium})$$

Thus,

$$\text{Eq'm in L.F. market} \Leftrightarrow \text{Eq'm in goods market}$$

Digression: Mastering models

To master a model, be sure to know:

1. Which of its variables are endogenous and which are exogenous.
2. For each curve in the diagram, know:
 - (a) definition
 - (b) intuition for slope
 - (c) all the things that can shift the curve
3. Use the model to analyze the effects of each item in 2c.

Mastering the loanable funds model (1 of 2)

Things that shift the saving curve:

- ▶ public saving
 - ▶ fiscal policy: changes in G or T
- ▶ private saving
 - ▶ preferences
 - ▶ tax laws that affect saving
 - ▶ 401(k)
 - ▶ IRA
 - ▶ replace income tax with consumption tax

CASE STUDY: The Reagan Deficits (1 of 2)

- ▶ Reagan policies during early 1980s:
 - ▶ increases in defense spending: $\Delta G > 0$
 - ▶ big tax cuts: $\Delta T < 0$
- ▶ Both policies reduce national saving:

$$\bar{S} = \bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G}$$

$$\uparrow \bar{G} \Rightarrow \downarrow \bar{S} \quad \downarrow \bar{T} \Rightarrow \uparrow C \Rightarrow \downarrow \bar{S}$$

▶ Numerical example



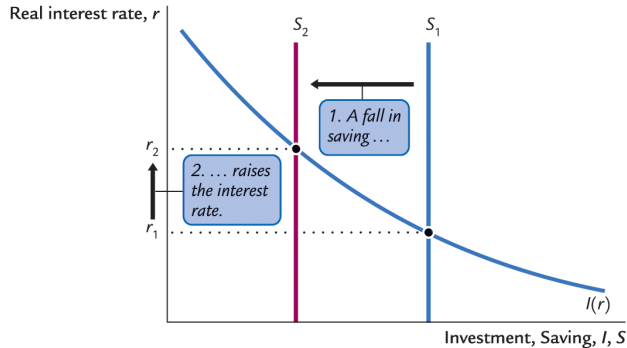
President Ronald Reagan,
1981 official portrait.

White House / Wikimedia, PD

CASE STUDY: The Reagan Deficits (2 of 2)

Saving decreases from S_1 to S_2 :

Results in the real interest rate increasing.



Are the data consistent with these results?

	1970s	1980s
$T - G$	-2.2	-3.9
S	19.6	17.4
r	1.1	6.3
I	19.9	19.4

$T - G$, S , and I are expressed as a percentage of GDP.

All figures are averages over the decade shown.

NOW YOU TRY: The effects of saving incentives

- ▶ Draw the diagram for the loanable funds model.
- ▶ Suppose the tax laws are altered to provide more incentives for private saving. (Assume that total tax revenue T does not change.)
- ▶ Does the Saving curve or the Investment curve shift?
- ▶ What happens to the interest rate and the level of savings/investment?

Mastering the loanable funds model (2 of 2)

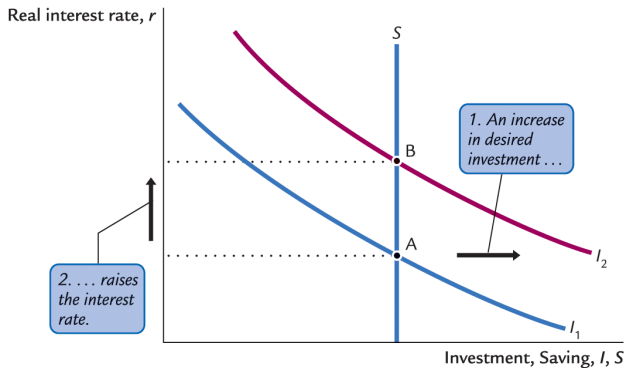
Things that shift the investment curve:

- ▶ some technological innovations
 - ▶ to take advantage of some innovations, firms must buy new investment goods
- ▶ tax laws that affect investment
 - ▶ example: investment tax credit

An increase in investment demand

Investment Demand increases from I_1 to I_2 :

Results in an increase in r . No change in equilibrium investment.



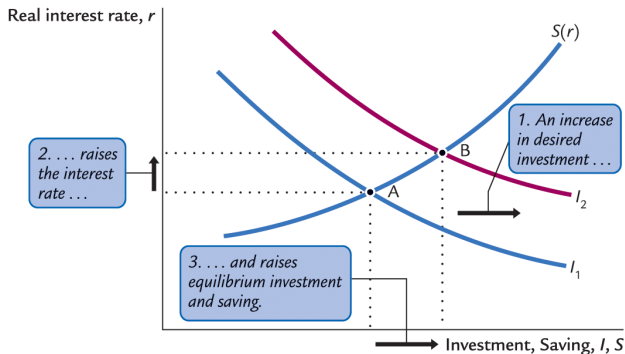
Saving and the interest rate

- ▶ Why might saving depend on r ?
- ▶ How would the results of an increase in investment demand be different?
 - ▶ Would r rise as much?
 - ▶ Would the equilibrium value of I change?

An increase in investment demand when saving depends on r

Investment Demand increases from I_1 to I_2 :

Results in an increase in r and an increase in equilibrium investment.



Chapter Summary, Part 1

- ▶ A closed economy's output is used for consumption, investment, and government spending.
- ▶ The real interest rate adjusts to equate the demand for and supply of:
 - ▶ goods and services.
 - ▶ loanable funds.

Chapter Summary, Part 2

- ▶ A decrease in national saving causes the interest rate to rise and investment to fall.
- ▶ An increase in investment demand causes the interest rate to rise but does not affect the equilibrium level of investment if the supply of loanable funds is fixed.

Appendix: Reagan deficits — a numerical worked example

Setup. A closed economy with

- ▶ Output: $\bar{Y} = 5,000$ (fixed at full employment)
- ▶ Consumption: $C = 200 + 0.75(Y - T)$
- ▶ Investment demand: $I(r) = 1,000 - 50r$ (with r in percent)
- ▶ Initial fiscal policy: $\bar{G} = 1,000$, $\bar{T} = 1,000$

Step 1 — initial equilibrium.

$$C = 200 + 0.75(5000 - 1000) = 3,200$$

$$\bar{S} = Y - C - G = 5000 - 3200 - 1000 = 800$$

$$I(r) = \bar{S}: \quad 1000 - 50r = 800 \implies r^* = 4\%$$

So initially $r^* = 4\%$ and $I^* = 800$.

◀ back

Appendix — Reagan shock 1: $\Delta G = +200$

Defense build-up: $\bar{G}$ rises from 1,000 to 1,200 (nothing else changes).

$$C = 3,200 \quad (\text{unchanged: } Y \text{ and } T \text{ fixed})$$

$$\bar{S} = 5000 - 3200 - 1200 = 600$$

$$1000 - 50r = 600 \implies r^* = 8\%$$

$$I^* = 600$$

$$\Delta G = +200 \Rightarrow \Delta S = -200 \Rightarrow \Delta r = +4\text{pp} \Rightarrow \Delta I = -200.$$

Higher government purchases crowd out private investment **dollar-for-dollar** because Y is fixed and C didn't change.

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Appendix — Reagan shock 2: $\Delta T = -250$

Tax cut: $\bar{T}$ falls from 1,000 to 750 (with $\bar{G}$ back at original 1,000).

$$Y - T = 5000 - 750 = 4,250$$

$$C = 200 + 0.75(4250) = 3,387.5$$

$$\bar{S} = 5000 - 3387.5 - 1000 = 612.5$$

$$1000 - 50r = 612.5 \implies r^* \approx 7.75\%$$

$$I^* \approx 612.5$$

Tax cut $\Delta T = -250$ raises disposable income by 250. With $MPC = 0.75$: $\Delta C = +187.5$, so $\Delta S = -187.5$, $\Delta r \approx +3.75\text{pp}$, $\Delta I \approx -187.5$.

Reagan combined the two shocks — both crowded out private investment via higher r .

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